

Streaming Live RAG - Architecture & Fixes Summary

1. Cost Control & LLM Dependency (LiteLLM + Gemini Flash)

Previously, there was a concern regarding hardcoded OpenAI dependencies causing unexpected costs. To resolve this, the system strictly utilizes 'LiteLLM', configured by default to use the free-tier Gemini 2.5 Flash ('gemini/gemini-2.5-flash'). All direct `openai.chat.completions` calls have been removed. This enforces cost control while retaining compatibility with OpenAI or Groq via environment variables.

2. Rigorous Metrics & Telemetry (`measure_metrics.py`)

A dedicated evaluation script ('`scripts/measure_metrics.py`') was created to move beyond gating checks. The system now computes:

- Recall@K against a gold standard passage set.
- Groundedness (using LLM-as-a-judge NLI/entailment).
- TTFT (Time To First Token) explicitly split into cold vs speculative hits.
- Cost per turn in USD, using LiteLLM's internal token cost trackers.

All telemetry (`telemetry/logger.py`) now records TTFT and Cost natively.

3. Delta Compute: 'Sharpen, Don't Restart'

The ambiguous 'late constraints' logic was completely rewritten. Instead of merely modifying the prompt and blindly sending old chunks back to the LLM, the system now implements active candidate filtering.

If a user says 'actually, only after 2020', the state manager extracts 'year_min = 2020'. The server then takes the existing candidate chunks and filters out any chunk whose metadata year is before 2020 or domain doesn't match. It then optionally re-sorts by recency before sending to the Synthesizer. This is computationally cheap and strictly enforces constraints without re-running vector fusion.

4. Environment Constraints & Scope

To adhere to the rapid prototype scope, Docker and docker-compose configurations were completely removed. Additionally, a Windows WDAC (Application Control) policy was blocking the SciPy DLLs inside the OneDrive folder. To bypass this, the Retriever was rewritten to use 'transformers' and 'torch' directly, eliminating the dependency on 'sentence_transformers' and 'scipy' entirely.

5. Corpus Limitations & Baseline

The mock dataset size (40 chunks across AI, Climate, Space, and Medicine) was retained, but explicit disclaimers were added to the README and the metrics script. Recall saturation on 40 documents is expected to be high, so the metrics act as a sanity check for the RRF fusion and decomposition pipelines, not a rigorous academic benchmark.